



COMP47350: Data Analytics (Conv)

Dr. Georgiana Ifrim

georgiana.ifrim@ucd.ie

Insight Centre for Data Analytics

ML-Labs Centre for Research Training in Machine Learning

School of Computer Science

University College Dublin

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Module Topics

- **Python Environment** (Anaconda, Python, Jupyter Notebook)

CRISP-DM Methodology for DA Projects

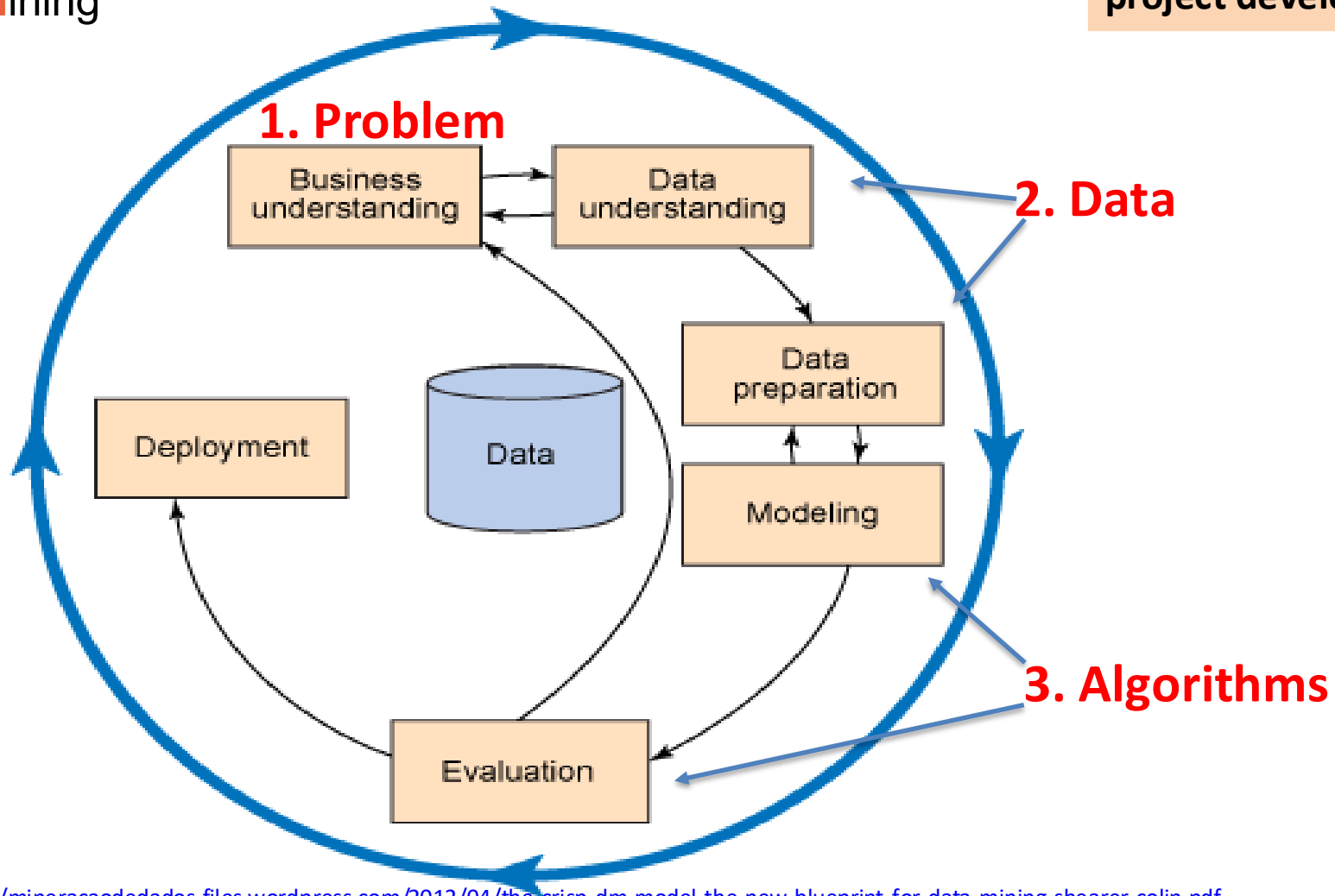
- **Problem: Problem Formulation**
- **Data: Getting Data** (Web scrapping, APIs, DBs)
- **Data: Understanding Data** (slicing, visualisation)
- **Data: Preparing Data** (cleaning, transformation)
- **Algorithm: Modeling & Evaluation** (machine learning)

Data Analytics Project Lifecycle:

CRISP-DM

CRISP-DM: **C**Ross-**I**ndustry **S**tandard **P**rocess for **D**ata **M**ining

6 stages for DA project development



Data Analytics Project Lifecycle:

CRISP-DM

CRISP-DM: **C**Ross-**I**ndustry **S**tandard **P**rocess for **D**ata **M**ining

1. Problem

1. **Business Understanding:** What is the business **problem**?

2. Data

2. **Data Understanding:** What is the **data** required to solve the business problem?

3. **Data Preparation:** Where is the **data**, how should it be collected, transformed and stored?

3. Algorithms

4. **Modeling:** What data analytics **algorithms** should be used?

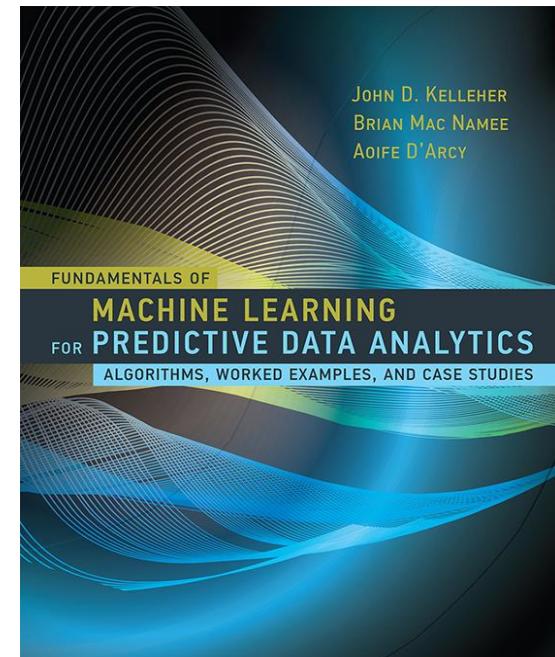
5. **Evaluation:** How well do the **algorithms** work?

6. **Deployment:** How can the analytics **results be integrated** into the work process (specific to the organisation)?

CRISP-DM: Business Understanding

- Converting the Business Problem into Analytics Solutions
- Assessing Feasibility
- **Case Study Problem:**
Motor Insurance Fraud Prediction

(lecture material follows the FMLPDA textbook shown in the image to the right, also see reference in the last slide)



CRISP-DM: Business Understanding

- **Converting the Business Problem into Analytics Solutions**
 - What is the business problem?
 - What are the goals that the business wants to achieve?
 - How does the business currently work?
 - In what ways could a predictive analytics model help to address the business problem?
 - Describe the predictive model, how will it be used, how will it help solve the original business problem, **what is the metric for success** (high model accuracy, increase in sales, decrease in operation cost)?

Case Study

Motor Insurance Fraud

Despite having a fraud investigation team that investigates up to 30% of all claims made, a **motor insurance company is still losing money due to fraudulent claims.**

- **Business entity:** motor insurance company
- **Problem:** fraudulent claims impact company revenue
- **Current solution:** fraud detection team investigates a sample of all the claims made (30% of all claims)
- **Improved solution:** use data available about past claims and their outcome (fraud/not-fraud) to focus the investigations team on a more relevant sample to investigate. **An algorithm can order the claims by the probability of fraud.**

Case Study

- What predictive analytics solutions could be proposed to help address this business problem?
- Potential solutions:
 - **Claim prediction: predict the probability of the outcome *fraud/not-fraud* for a given claim**
 - Member prediction
 - Application prediction
 - Payment prediction

CRISP-DM: Business Understanding

- **Assessing Feasibility**

- **Data:** Is the data required by this solution available, or could it be made available?
- **Capacity:** What is the capacity of the business to utilize the insights that the analytics solution will provide? (e.g., integration: how does the analytics solution fit into the existing business process?)
- **Impact:** How do we know the solution is successful? (e.g., minimum model accuracy, impact on business operations, what is a good baseline to compare against?)

Case Study: Claim Prediction

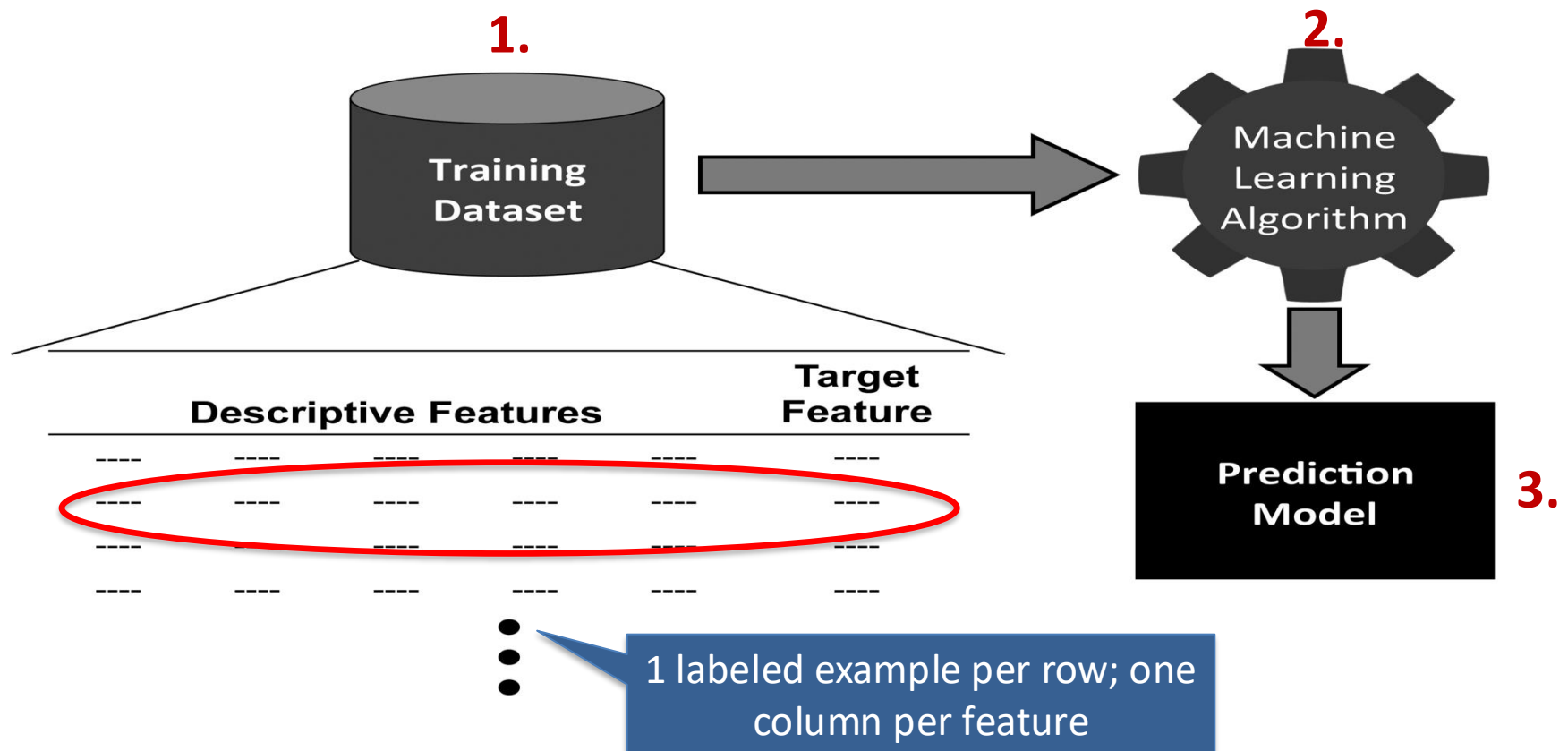
- What are the **data** and **capacity** requirements for the proposed **Claim Prediction** analytics solution?
- **Data Requirements:**
 - INPUT: Details of each claim and claimant, the related policy, other info
 - OUTPUT: past claims marked as **fraudulent** and **not-fraudulent** (supervised label)
- **Capacity Requirements:** A way to inform claim investigators that some claims are prioritized above others. Information about claims should become available in a timely manner (features and predicted outcome) so that the claims investigation process is not delayed by the model.
- **Impact?** How do we know the solution is **successful**? (**Discussion**)
 - How can it be used?
 - What could go wrong?

CRISP-DM: Data Understanding

- **Designing the Analytics Base Table** (e.g., the CSV file describing examples via **descriptive features**)
- **Designing & Implementing Features**
 - Different Types of Data
 - Different Types of Features
 - Handling Time
 - Legal Issues
 - Implementing Features

Analytics Base Table (CSV)

The basic structure in which we capture historical datasets is the **analytics base table** (e.g., a spreadsheet). Ideally, it comes with a **data dictionary**: a file defining and describing each feature.



Analytics Base Table

Motor Insurance Claim Prediction (target OUTPUT: **Fraud**/**Not-fraud**)

ID	TYPE	INC.	MARITAL STATUS	NUM. CLMNTS.	INJURY TYPE	HOSPITAL STAY	CLAIM AMT.
1	CI	0	Married	2	Soft Tissue	No	1 625
2	CI	0		2	Back	Yes	15 028
3	CI	54 613		1	Broken Limb	No	-9 999
4	CI	0		3	Serious	Yes	270 200
			⋮			⋮	

ID	TOTAL CLAIMED	NUM. CLAIMS	NUM. CLAIMS 3 MONTHS	AVG. CLAIMS PER YEAR	AVG. CLAIMS RATIO	NUM. SOFT TISSUE	% SOFT TISSUE
1	3 250	2	0	1	1	2	1
2	60 112	1	0	1	1	0	0
3	0	0	0	0	0	0	0
4	0				0		
					⋮		

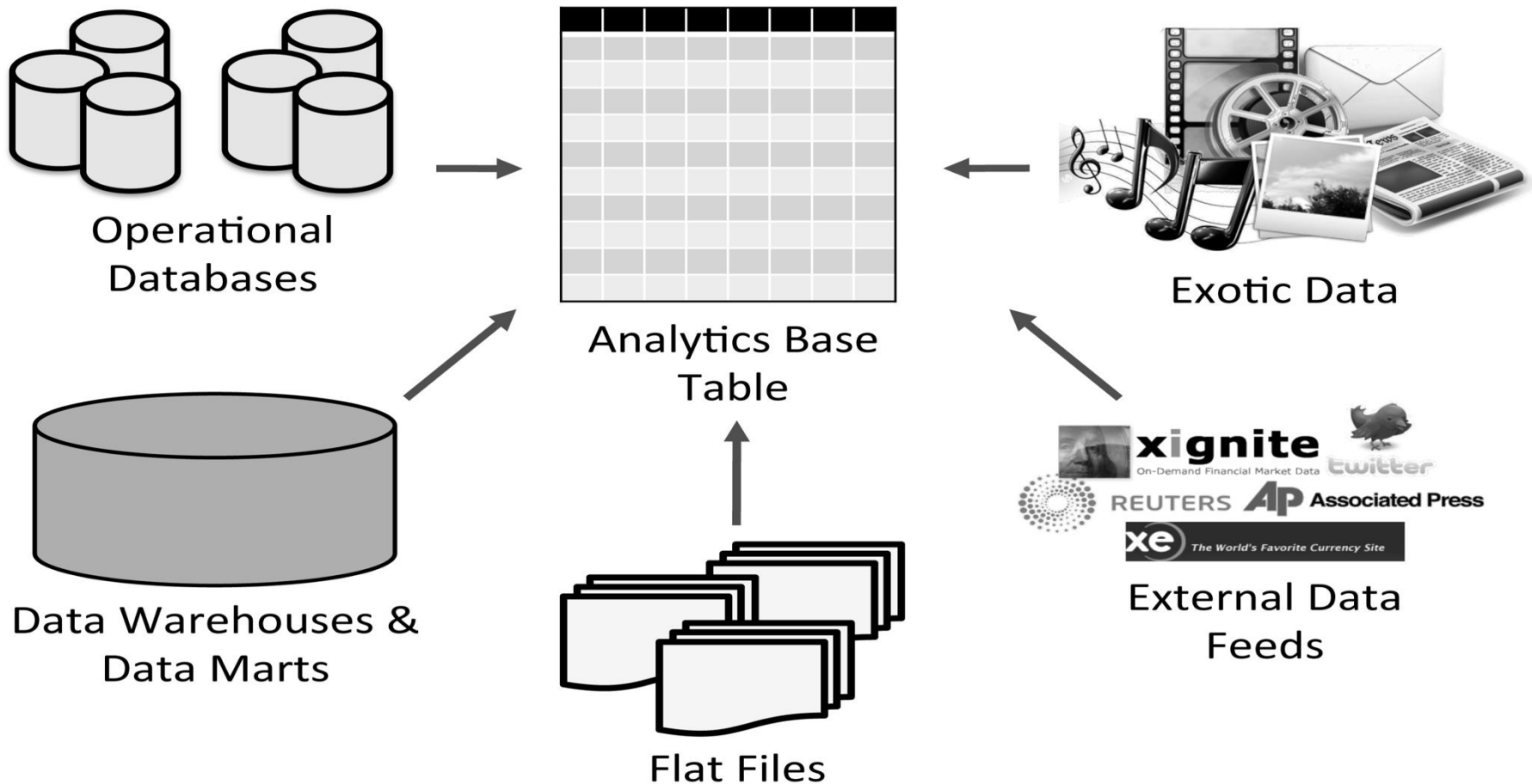
Descriptive Features

Target Feature

ID	UNSUCC. CLAIMS	CLAIM AMT. REC.	CLAIM DIV.	CLAIM TO PREM.	REGION	FRAUD FLAG
1	2	0	0	32.5	MN	1
2	0	15 028	0	57.14	DL	0
3	0	572	0	-89.27	WAT	0
4	0	270 200	0	30.186	DL	0
					⋮	

Analytics Base Table

Many data sources are typically combined to create an analytics base table (next lecture: hands-on data gathering)

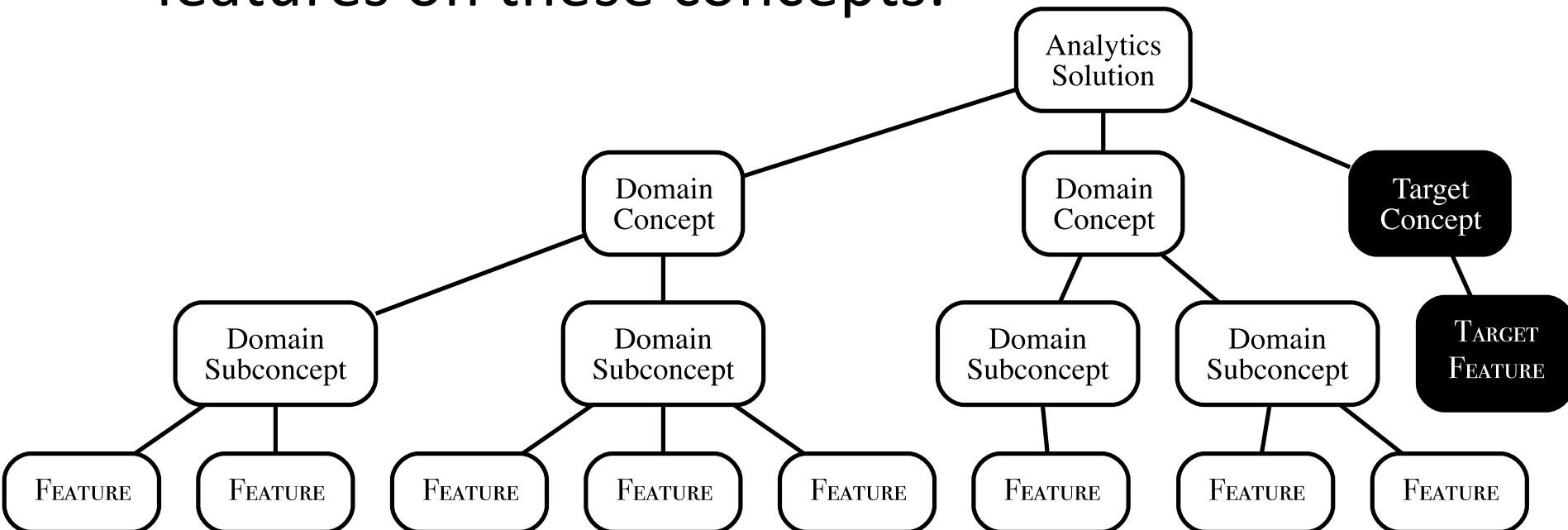


Analytics Base Table

- The **prediction subject** defines the basic level at which predictions are made (e.g., **claim prediction**)
- Each row in the ABT will represent one instance of the prediction subject—the phrase **one-row-per-subject** is often used to describe this structure
- Each row/example in an ABT is composed of a set of **descriptive features** and a **target feature**

Analytics Base Table

- **Defining descriptive features can be difficult**
- A good way to define features is to identify the key **domain concepts** and then to base the features on these concepts.



Analytics Base Table

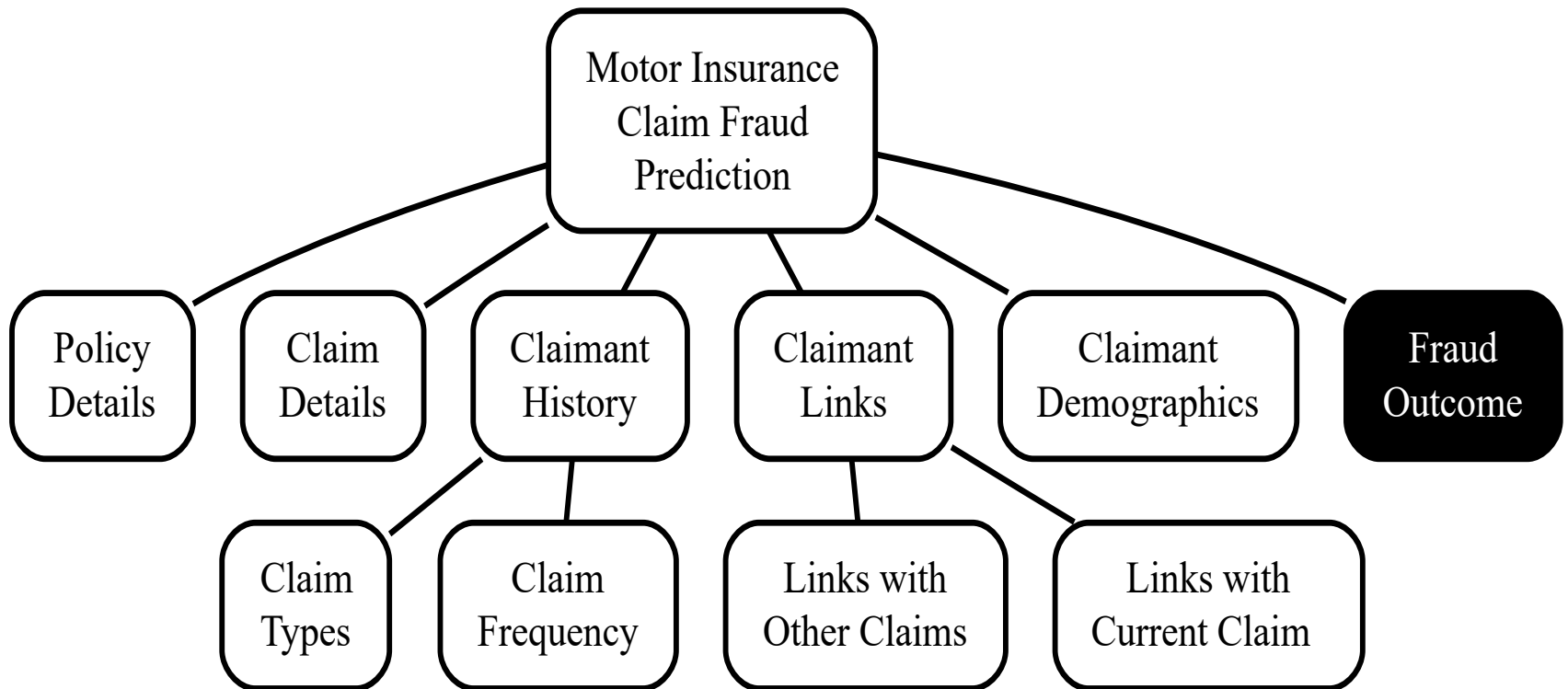
- General **domain concepts** that are often useful:

Prediction Subject Details (claimant, claim details)

- Applicant Demographics (age, country, education)
- Usage
- Changes in Usage
- Other?

Analytics Base Table

Domain concepts for **motor insurance claim prediction**



Types of Features

Features (i.e., columns in the ABT) can be of type:

- Numeric
- Binary
- Ordinal
- Interval
- Date
- Categorical
- Textual

We typically convert all the features into numeric features.
Most learning algorithms expect numeric inputs.

Types of Features/Data

ID	NAME	DATE OF BIRTH	GENDER	CREDIT RATING	COUNTRY	SALARY
0034	Brian	22/05/78	male	aa	ireland	67,000
0175	Mary	04/06/45	female	c	france	65,000
0456	Sinead	29/02/82	female	b	ireland	112,000
0687	Paul	11/11/67	male	a	usa	34,000
0982	Donald	01/12/75	male	b	australia	88,000
1103	Agnes	17/09/76	female	aa	sweden	154,000

Ordinal? (points to ID and CREDIT RATING)

Categorical? (points to COUNTRY)

Textual? (points to NAME)

Interval? (points to DATE OF BIRTH)

Binary? (points to GENDER)

Numeric? (points to SALARY)

References

- **The CRISP-DM Model: The New Blueprint for Data Mining**, C. Shearer, Journal of Data Warehousing, 2000

<https://mineracaodedados.files.wordpress.com/2012/04/the-crisp-dm-model-the-new-blueprint-for-data-mining-shearer-colin.pdf>

- **FMLPDA Book: Fundamentals of Machine Learning for Predictive Data Analytics**, John D. Kelleher, Brian Mac Namee, Aoife D'Arcy, MIT Press, 2015

<https://mitpress.mit.edu/books/fundamentals-machine-learning-predictive-data-analytics>

<http://machinelearningbook.com>

- Web resources:

<http://www.kdnuggets.com/2015/11/seven-steps-machine-learning-python.html>